

The Universal Nonlinear Deformation Law: Derivation from Relativistic Corrections and Extension to All Energy Scales

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Abstract

The Active Vacuum Framework (AVF) identifies the vacuum as the sole dynamical entity, carrying a Poincaré-invariant ambient energy density at each energy scale. Mass depletes this energy density locally, creating a deformation field and a consequent compactness parameter. In the weak-field limit the local energy density follows the linear law. This paper derives the appropriate nonlinear deformation law from first principles by incorporating the relevant relativistic corrections: (i) gravitational redshift of the vacuum energy and (ii) spatial metric stretching of the escape path.

The scale-independent horizon condition $\mathcal{C}(r_h) = 1/2$, at which the effective energy density vanishes, is derived as the physical criterion for complete vacuum depletion at any scale. When applied to the gravitational scale this gives the Schwarzschild radius $r_S = 2GM/c^2$ exactly. Applied to the electromagnetic, QCD, Higgs, and electroweak scales, the same law gives the characteristic horizon radii of those interactions. The five-scale hierarchy is established, and its consequences in terms of force strengths and horizon radii are derived. The QCD-scale prediction for the proton confinement radius agrees with the muonic hydrogen measurement to within 0.2%.

1 Introduction

The formulation of the Active Vacuum Framework (AVF) started with five postulates in congruence with basic observations in nature and three key physical variables [3]. The primary observation is that any mass M draws energy from the all-pervading Poincaré-invariant ambient vacuum energy in its surroundings, creating a deformation field

$$\mathcal{D}_G(r) = \mathcal{A} \cdot \frac{GM}{c^2 r}, \quad (1)$$

so that the local ambient energy density becomes $\mathcal{H}_{G,N}(r) = \mathcal{A}(1 - GM/c^2 r)$. The compactness parameter $\mathcal{C}_G(r) = \mathcal{D}_G/\mathcal{A} = GM/(c^2 r)$ measures the fractional depth of the deformation at radius r from the source mass M . A small value of the compactness parameter $\mathcal{C}_G(r)$ means

the vacuum is barely affected; a value approaching $1/2$ means the vacuum energy has been reduced to half its ambient level. Newton's law of gravitation was derived as the gradient of $\mathcal{D}_G$.

The nonlinear deformation law is also stated in [3] as an extension of the linear deformation given by (1):

$$\mathcal{H}_S^{\text{eff}}(r) = \mathcal{H}_S^{(0)} \sqrt{1 - 2\mathcal{C}_S(r)}, \quad (2)$$

with a physical motivation but without a full derivation. The motivation was that the linear law $\mathcal{H}_G = \mathcal{A}(1 - \mathcal{C}_G)$ assumes the undisturbed background $\mathcal{A}$ everywhere, which overestimates the available vacuum energy near a massive source. The correct nonlinear form follows when two relativistic effects that become significant near compact objects are accounted for.

The purpose of this paper is to carry out that derivation explicitly and step by step, so that the generic nonlinear relation given by Eq. (2) is established from physical principles rather than assumed. The derivation is self-contained. It requires only the definition of the deformation field, the two relativistic corrections explained in Section 4, and the principle that vacuum energy is gravitationally redshifted at the same rate as all other energy. No prior knowledge of general relativity is assumed: the Schwarzschild metric emerges from the derivation, rather than being an input to it.

The paper then extends the nonlinear law to all five energy scales of AVF. The horizon condition $\mathcal{C}_S(r_h) = 1/2$ gives a characteristic radius r_h at each scale: the Schwarzschild radius at the gravitational scale, the classical electron radius at the electromagnetic scale, the proton confinement radius at the QCD scale, and corresponding radii at the Higgs and electroweak scales. The same mathematical law governs vacuum depletion from cosmological distances down to sub-nuclear scales.

The paper is organised as follows. Section 2 reviews the linear profile and identifies its limitations. Section 4 derives the two relativistic corrections and combines them into the nonlinear law. Section 5 derives the horizon condition and applies it to the gravitational scale. Section 6 extends the law to all five energy scales and derives the scale hierarchy and horizon radii. Section 7 summarises the developments in this paper.

2 Limitation of Newtonian Deformation Profile

We state the linear Newtonian energy profile developed in the companion Newtonian paper [4], and show its limitations by a thorough analysis. Note that we have assumed a static mass distribution so that the kinetic energy density at any point $\mathcal{K}(r)$ is assumed to be zero. It could be applicable for a dynamic scenario when motion is slow, and slowness is quantified here.

2.1 The Linear Profile

The gravitational deformation field [4] around a point mass M at the origin is given by $\mathcal{D}_G(r) = \mathcal{A} \cdot GM/(c^2 r)$. Substituting into $\mathcal{H}_G(r) = \mathcal{A} - \mathcal{D}_G(r)$, the local ambient energy density in the Newtonian approximation is:

$$\mathcal{H}_{G,N}(r) = \mathcal{A} - \mathcal{D}_G(r) = \mathcal{A} \left(1 - \frac{GM}{c^2 r} \right) = \mathcal{A}(1 - \mathcal{C}_G(r)). \quad (3)$$

This is the linear deformation profile. The word “linear” refers to the fact that the depletion $\mathcal{D}_G(r)/\mathcal{A} = GM/(c^2r)$ is a linear function of the compactness parameter $\mathcal{C}_G$: doubling M doubles the depletion at any radius, and halving r doubles it again.

The linear profile holds whenever $\mathcal{C}_G(r) \ll 1$, which covers all of everyday gravitational physics. To appreciate how small the depletion is in ordinary circumstances, we can evaluate $\mathcal{C}_G$ at two familiar locations:

$$\text{Earth's surface: } \mathcal{C}_G(R_\oplus) = \frac{GM_\oplus}{c^2 R_\oplus} \approx \frac{6.674 \times 10^{-11} \times 5.97 \times 10^{24}}{(3 \times 10^8)^2 \times 6.37 \times 10^6} \approx 7 \times 10^{-10}, \quad (4)$$

$$\text{Sun's surface: } \mathcal{C}_G(R_\odot) = \frac{GM_\odot}{c^2 R_\odot} \approx \frac{6.674 \times 10^{-11} \times 1.99 \times 10^{30}}{(3 \times 10^8)^2 \times 6.96 \times 10^8} \approx 2 \times 10^{-6}. \quad (5)$$

In both cases $\mathcal{C}_G \ll 1$: the vacuum is barely affected by the presence of these masses, and the linear approximation is excellent. The compactness parameter reaches order unity only for extremely dense objects such as neutron stars ($\mathcal{C}_G \sim 0.2$) and near the Schwarzschild radius of a black hole ($\mathcal{C}_G \rightarrow 1/2$).

2.2 The Zero-Crossing of the Linear Profile

Setting $\mathcal{H}_{G,N}(r_*^N) = 0$ and solving for r_*^N :

$$1 - \frac{GM}{c^2 r_*^N} = 0 \implies r_*^N = \frac{GM}{c^2} = \frac{r_S}{2}, \quad (6)$$

where $r_S \equiv 2GM/c^2$ is the Schwarzschild radius. The linear profile predicts that the local ambient energy density reaches zero at half the Schwarzschild radius. This is the Newtonian estimate for the boundary of complete vacuum depletion: the radius inside which, according to the linear approximation, the vacuum has given up all of its energy to the deformation.

As shown in the context of the estimation of the black hole horizon [5], the correct zero-crossing occurs at r_S , not $r_S/2$. The linear profile is therefore required to be modified by a factor of 2. The source of this discrepancy is explained in the next subsection.

2.3 Linear Profile is an Approximate Description

The linear profile was derived under the assumption that the source reduces the ambient energy density by the amount $\mathcal{D}_G(r) = \mathcal{A} \cdot GM/(c^2r)$ at each point, calculated against the background $\mathcal{A}$ as a fixed, undisturbed reference. This is the same assumption made in Newtonian gravity: the gravitational potential $\Phi(r) = -GM/r$ is calculated as if the background space is flat and undistorted.

This assumption has a physical cost. It ignores two effects that become significant as $\mathcal{C}_G$ approaches $1/2$. Both act in the same direction: they increase the effective depletion of the vacuum energy beyond the Newtonian estimate, pushing the zero-crossing from $r_S/2$ outward to r_S . To understand them intuitively, imagine asking the following question: if a signal (a photon, a gravitational influence, or a vacuum excitation) is emitted at radius r and tries to climb outward to infinity, how much energy does it need? The Newtonian answer is $\mathcal{D}_G(r)$. The correct relativistic answer is larger due to the following two reasons.

Reason 1: Gravitational redshift: A photon emitted at radius r with local energy E_{loc} arrives at infinity with a smaller energy $E_{\infty} = E_{\text{loc}}\sqrt{1 - r_S/r}$. The energy has been redshifted by the factor $\sqrt{1 - r_S/r} < 1$. This means that the vacuum energy at radius r , which drives the outward escape, is also worth less at infinity than its local value suggests. The effective vacuum energy available to support escape is reduced by the same redshift factor.

Reason 2: Spatial metric stretching: In curved spacetime near a massive object, the actual physical distance between two points at radii r and $r + dr$ is $dr_{\text{prop}} = dr/\sqrt{1 - r_S/r} > dr$. The ruler is stretched. This means the escape path from r to infinity is physically longer than the coordinate distance, and the energy required to traverse it is correspondingly larger. The Newtonian calculation, which uses the coordinate distance, underestimates the escape work.

Both effects become larger as $r \rightarrow r_S$ from outside, because $\sqrt{1 - r_S/r} \rightarrow 0$ as $r \rightarrow r_S$. Near the Schwarzschild radius, the linear profile fails completely. The next section derives the exact relativistic profile that accounts for both effects simultaneously.

3 Two Relativistic Corrections

Consider a photon (or any signal) emitted at radius r and attempting to escape to infinity. What energy does it carry to a far observer? The linear profile says the photon simply carries energy reduced by $\mathcal{D}_G(r)/\mathcal{A} = GM/(c^2r)$. Therefore, we apply the following two corrections to arrive at the appropriate formulation.

Correction 1: Gravitational redshift: Energy measured locally at radius r is not the same as energy received by a distant observer. A photon climbing out of a gravitational well loses energy. If a photon is emitted with local energy E_{loc} at radius r , it arrives at infinity with energy:

$$E_{\infty} = E_{\text{loc}} \times \sqrt{1 - \frac{r_S}{r}}. \quad (7)$$

The factor $\sqrt{1 - r_S/r}$ is the gravitational redshift factor, a direct consequence of energy conservation in a gravitational field. It is less than 1 for all $r > r_S$ and goes to zero as $r \rightarrow r_S$.

The same redshift applies to the ambient vacuum energy. The ambient vacuum energy density is a form of energy, and it is subject to gravitational redshift just as any other form of energy is. The local ambient energy density $\mathcal{A}$ at radius r , as measured by a distant observer, appears reduced by the same redshift factor:

$$\mathcal{A}^{\text{eff}}(r) = \mathcal{A} \times \sqrt{1 - \frac{r_S}{r}}. \quad (8)$$

Where time flows slower (closer to the source), all energy processes slow and all energy densities appear reduced to the outside observer. The ambient vacuum energy participates in this bookkeeping equally with matter and radiation.

Correction 2: Spatial metric stretching: The second correction concerns the path itself. When measuring distances near a massive body, the radial distance is not simply the

coordinate difference dr . The proper (physically measured) radial distance element is:

$$ds_{\text{prop}} = \frac{dr}{\sqrt{1 - r_S/r}}. \quad (9)$$

Near the Schwarzschild radius, where $r_S/r \rightarrow 1$, the proper distance becomes much larger than the coordinate distance. A signal escaping from r to infinity must traverse a longer actual path than the coordinate integral suggests. This means the effective depletion, the vacuum energy consumed in the escape, is greater than the Newtonian estimate.

The two corrections together replace r_S/r in the linear profile with $r_S/r + r_S/r = 2GM/(c^2 r) = 2\mathcal{C}_G$. More precisely, the combined effect of redshift and path stretching replaces the linear fractional depletion $\mathcal{C}_G$ with the nonlinear form $1 - \sqrt{1 - 2\mathcal{C}_G}$. The derivation of this replacement is carried out explicitly in the next section.

4 The Relativistic Deformation Profile

The effective depletion fraction: The total fractional depletion that a signal must overcome to escape from radius r to infinity, accounting for both relativistic corrections of Section 3, is:

$$\frac{\mathcal{D}_G^{\text{eff}}(r)}{\mathcal{A}} = 1 - \sqrt{1 - \frac{r_S}{r}} = 1 - \sqrt{1 - 2\mathcal{C}_G(r)}. \quad (10)$$

In the weak-field limit $\mathcal{C}_G \ll 1$, expanding the square root using $\sqrt{1 - x} \approx 1 - x/2$ for small x :

$$1 - \sqrt{1 - 2\mathcal{C}_G} \approx \mathcal{C}_G = \frac{GM}{c^2 r}, \quad (11)$$

recovering the Newtonian result (3) exactly. The relativistic formula therefore contains the Newtonian one as its limiting case.

Origin of $(1 - \sqrt{1 - 2\mathcal{C}_G})$: The origin of this depletion form can be understood directly from the following discussion. Note that the depletion fraction is the fraction of $\mathcal{A}$ that a signal uses up escaping from r to infinity. Let us define $f(r)$ as the fraction that *remains* after the escape: $f = 1 - \text{depletion fraction}$. The following two conditions determine f :

- (i) $f \rightarrow 1$ as $r \rightarrow \infty$: far from the source, no depletion, full ambient energy available.
- (ii) $f \rightarrow 0$ at the horizon r_S : at the horizon, the entire ambient energy is consumed in the escape attempt. Nothing remains to carry the signal outward.

The Schwarzschild geometry, which is the geometry of empty space around a spherical mass, determines that $f(r) = \sqrt{1 - r_S/r} = \sqrt{g_{00}(r)}$. This is not an assumption: it is what the geometry of curved spacetime requires when these two conditions are imposed on a function that must also satisfy the Einstein field equations in vacuum.

The effective local ambient energy density: The effective ambient energy density that remains at radius r after accounting for both relativistic corrections is:

$$\mathcal{H}_G^{\text{eff}}(r) = \mathcal{A} - \mathcal{D}_G^{\text{eff}}(r) = \mathcal{A} \left[1 - \left(1 - \sqrt{1 - 2\mathcal{C}_G} \right) \right] = \mathcal{A} \sqrt{1 - 2\mathcal{C}_G(r)}. \quad (12)$$

This is the relativistic deformation profile. It replaces the linear Newtonian profile (3) when relativistic effects are important and vice versa.

Remark 1 (Poincaré invariance of $\mathcal{A}$ and the meaning of $\mathcal{H}_G^{\text{eff}}$). *A careful reader may ask: if the ambient vacuum energy density $\mathcal{A}$ is Poincaré invariant and therefore the same for all observers, how can $\mathcal{H}_G^{\text{eff}}(r) = \mathcal{A}\sqrt{1 - 2\mathcal{C}_G(r)}$ depend on position? The answer resolves an important distinction between two different quantities.*

The ambient energy density $\mathcal{A}$ is indeed the same for all inertial observers in the undisturbed vacuum. It is invariant under translations, rotations, and Lorentz boosts: an observer at rest and an observer moving at constant velocity in empty flat space both measure $\mathcal{A}$. This invariance is not violated by the nonlinear law.

What $\mathcal{H}_G^{\text{eff}}(r)$ describes is not the local value of $\mathcal{A}$ at position r . It is the fraction of the local vacuum energy at r that is accessible to a distant observer: the energy that can actually reach infinity after climbing out of the gravitational potential well. The factor $\sqrt{1 - 2\mathcal{C}_G(r)} = \sqrt{g_{00}(r)}$ is the gravitational redshift conversion between local energy and remotely received energy. It accounts for what is lost in transit, not for any change in the intrinsic vacuum energy density.

To state this precisely: the local observer at radius r measures an ambient vacuum energy density of $\mathcal{A} - \mathcal{D}_G(r)$ in the deformed vacuum, not $\mathcal{H}_G^{\text{eff}}(r)$. A distant observer asking “how much of that local energy can reach me?” receives the answer $\mathcal{H}_G^{\text{eff}}(r) = \mathcal{A}\sqrt{g_{00}}$. The deformation $\mathcal{D}_G$ is a real, local reduction of the vacuum energy caused by the mass. The additional factor $\sqrt{g_{00}}$ is the redshift: energy tax paid on the journey outward. Poincaré invariance of $\mathcal{A}$ refers to the undisturbed vacuum background; the deformation and its redshift are both consequences of the presence of the mass, not violations of the background symmetry.

Verification of the two limits: For $\mathcal{C}_G \ll 1$: $\sqrt{1 - 2\mathcal{C}_G} \approx 1 - \mathcal{C}_G$, so $\mathcal{H}_G^{\text{eff}} \approx \mathcal{A}(1 - \mathcal{C}_G)$, which is exactly the linear Newtonian profile. The relativistic profile contains the Newtonian one. At $\mathcal{C}_G = 1/2$: $\sqrt{1 - 2 \times 1/2} = \sqrt{0} = 0$, so $\mathcal{H}_G^{\text{eff}} = 0$. The effective ambient energy density vanishes at the radius where the compactness equals one half. The condition $\mathcal{C}_G = 1/2$ therefore defines the horizon.

Radius	$\mathcal{C}_G(r)$	$\mathcal{H}_{G,N}/\mathcal{A}$	$\mathcal{H}_G^{\text{eff}}/\mathcal{A}$
$r \rightarrow \infty$	0	1	1
$r = 4r_S$	1/8	0.875	0.866
$r = 2r_S$	1/4	0.750	0.707
$r = r_S$	1/2	0.500	0
$r = r_S/2$	1	0	imaginary

The table shows the crucial difference: the linear profile reaches zero at $r = r_S/2$ (compactness = 1), but the relativistic profile reaches zero at $r = r_S$ (compactness = 1/2). The Newtonian estimate places the horizon at half the correct radius. The factor of two comes from the square-root structure of the relativistic correction, which exactly doubles the effective depletion compared to the Newtonian estimate at the horizon.

5 The Horizon Condition

Derivation of the Schwarzschild radius: The horizon is the theoretical boundary of the body with radius r_S (assuming the body at the origin) at which the effective ambient energy density reaches zero:

$$\mathcal{H}_G^{\text{eff}}(r_S) = 0 \implies \sqrt{1 - 2\mathcal{C}_G(r_S)} = 0 \implies \mathcal{C}_G(r_S) = \frac{1}{2}. \quad (13)$$

Substituting $\mathcal{C}_G(r) = GM/(c^2 r)$:

$$\frac{GM}{c^2 r_S} = \frac{1}{2} \implies r_S = \frac{2GM}{c^2}. \quad (14)$$

This is the Schwarzschild radius, obtained here as the radius at which the compactness equals one half and the effective ambient energy density is completely exhausted.

Physical meaning of the horizon: At $r = r_S$, the effective ambient energy density $\mathcal{H}_G^{\text{eff}}$ vanishes. The entire ambient vacuum energy at the gravitational scale has been consumed by the relativistic depletion at this radius. A photon, or any other signal, attempting to move outward from r_S finds no ambient energy density to support its propagation. It cannot escape. The horizon does not imply a force barrier; rather it is the bounding surface of a region where the effective ambient vacuum energy at the gravitational scale is fully exhausted. A photon at r_S has non-zero velocity and non-zero local energy. What it lacks is not energy but the effective ambient energy that would carry the electromagnetic wave outward. This is the physical mechanism of the black hole horizon in the AVF picture.

The metric as the squared energy ratio: From Eq. (12):

$$\frac{\mathcal{H}_G^{\text{eff}}(r)}{\mathcal{A}} = \sqrt{1 - \frac{r_S}{r}}, \quad \text{so} \quad \left(\frac{\mathcal{H}_G^{\text{eff}}(r)}{\mathcal{A}} \right)^2 = 1 - \frac{r_S}{r} = g_{00}(r). \quad (15)$$

The Schwarzschild metric component $g_{00}(r)$ is the squared ratio of the effective local ambient energy density to the undisturbed background. Spacetime curvature in the AVF picture is the macroscopic expression of the effective ambient energy density profile at the gravitational scale, as established in the companion EFE paper [6].

6 Extension to All Energy Scales

The derivation of Section 4 used the following:

- (i) A deformation field $\mathcal{D}_G(r)$ that depletes the ambient energy density at each point.
- (ii) A compactness $\mathcal{C}_G(r) = \mathcal{D}_G/\mathcal{A}$ measuring the fractional depletion.
- (iii) Two relativistic corrections (redshift and path stretching) that replace the linear depletion fraction $\mathcal{C}_G$ with the nonlinear form $1 - \sqrt{1 - 2\mathcal{C}_G}$.

Each of these three ingredients is applicable at every energy scale, not only the gravitational one. At each scale S , there is an ambient energy density $\mathcal{H}_S^{(0)}$, a deformation field $\mathcal{D}_S(r)$ produced by the source at that scale, and a compactness $\mathcal{C}_S(r) = \mathcal{D}_S/\mathcal{H}_S^{(0)}$. The two relativistic corrections apply equally: the ambient energy at scale S is redshifted at the same rate as all other energy at that scale, and the propagation path is stretched by the same geometric factor.

6.1 The Universal Nonlinear Deformation Law

Replacing $G \rightarrow S$, $\mathcal{A} \rightarrow \mathcal{H}_S^{(0)}$, and $\mathcal{C}_G \rightarrow \mathcal{C}_S$ in Eq. (12):

$$\mathcal{H}_S^{\text{eff}}(r) = \mathcal{H}_S^{(0)} \sqrt{1 - 2\mathcal{C}_S(r)}, \quad \mathcal{C}_S(r) = \frac{\mathcal{D}_S(r)}{\mathcal{H}_S^{(0)}}. \quad (16)$$

This is the universal nonlinear deformation law, which governs the effective ambient energy density at every scale S in the presence of a source of compactness $\mathcal{C}_S(r)$.

The universal horizon condition: At every scale, the horizon is where the effective ambient energy density is fully exhausted:

$$\mathcal{H}_S^{\text{eff}}(r_*) = 0 \iff \mathcal{C}_S(r_*) = \frac{1}{2}. \quad (17)$$

Different scales have different compactness parameters $\mathcal{C}_S(r)$, and therefore different horizon radii. The horizon condition itself is the same for all scales.

Limiting forms: The universal law has two limiting forms. For $\mathcal{C}_S(r) \ll 1$ (weak deformation):

$$\mathcal{H}_S^{\text{eff}}(r) \approx \mathcal{H}_S^{(0)}(1 - \mathcal{C}_S(r)), \quad (18)$$

which is the linear deformation law of the postulates paper [3]. For $\mathcal{C}_S(r) \rightarrow 1/2$ (approach to horizon):

$$\mathcal{H}_S^{\text{eff}}(r) \rightarrow \mathcal{H}_S^{(0)} \sqrt{1 - 2\mathcal{C}_S} \rightarrow 0, \quad (19)$$

the effective ambient energy density vanishes and the horizon forms. The force at scale S is proportional to $\nabla \mathcal{D}_S$. In the weak-field limit it gives the classical long-range force at that scale. Near the horizon it gives the strongly nonlinear confinement behaviour.

6.2 The Compactness Parameter and Horizon Radius at Each Scale

The universal law (16) takes a specific form at each scale once the compactness parameter $\mathcal{C}_S$ is identified. The compactness is determined by the source and the coupling at that scale, by analogy with the gravitational case where the source is mass M and the coupling is G .

Gravitational scale. Source: mass M . Coupling: Newton's constant G .

$$\mathcal{C}_G(r) = \frac{GM}{c^2 r}. \quad (20)$$

Horizon condition $\mathcal{C}_G(r_S) = 1/2$ gives the Schwarzschild radius $r_S = 2GM/c^2$.

Electromagnetic scale. Source: electric charge e . Coupling: fine-structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$. The Coulomb potential $-\alpha\hbar c/r$ defines the compactness by analogy with the gravitational case ($GM \leftrightarrow \alpha\hbar c/m_e$):

$$\mathcal{C}_E(r) = \frac{\alpha\hbar c}{\mathcal{H}_E^{(0)} \cdot r} = \frac{e^2}{4\pi\epsilon_0 m_e c^2 r}. \quad (21)$$

Horizon condition $\mathcal{C}_E(r_e^{\text{cl}}) = 1/2$ gives the classical electron radius:

$$r_e^{\text{cl}} = \frac{e^2}{2\pi\epsilon_0 m_e c^2} \approx 2.82 \text{ fm}. \quad (22)$$

This is derived in detail in the companion Coulomb paper [7].

QCD scale. Source: colour charge. Coupling: strong coupling constant α_s . The one-gluon exchange potential $-4\alpha_s\hbar c/(3r)$ defines the compactness by analogy with the gravitational case ($GM \rightarrow 4\alpha_s\hbar c/3$):

$$\mathcal{C}_Q(r) = \frac{4\alpha_s}{3} \cdot \frac{\hbar c}{\Lambda_{\text{QCD}} r}. \quad (23)$$

Horizon condition $\mathcal{C}_Q(r_p) = 1/2$ gives the proton confinement radius:

$$r_p = \frac{8\alpha_s}{3} \cdot \frac{\hbar c}{\Lambda_{\text{QCD}}} \approx 0.843 \text{ fm}, \quad (24)$$

in 0.2% agreement with the muonic hydrogen measurement of the proton charge radius [1].

Higgs scale. Source: confined charge quantum $(-e/3)$ inside the electron structure. The Higgs domain wall at r_e^H marks the horizon of the Higgs-scale deformation. Pressure balance between the electromagnetic deformation and the Higgs surface tension gives [2]:

$$r_e^H = \frac{e^2}{8\pi\epsilon_0 m_e c^2} \approx 2.82 \text{ fm}, \quad (25)$$

consistent with the horizon condition $\mathcal{C}_H(r_e^H) = 1/2$ at the Higgs scale. The full derivation is given in the electron structure paper of this series [2].

Electroweak scale. Source: weak isospin charge. Coupling: weak coupling constant g_W . The characteristic length is the W boson Compton wavelength:

$$\mathcal{C}_W(r) = \frac{g_W^2 \hbar c}{\mathcal{H}_W^{(0)} \cdot r}, \quad r_W = \frac{\hbar}{m_W c} \approx 2.5 \times 10^{-3} \text{ fm}. \quad (26)$$

The electroweak horizon radius r_W is the Compton wavelength of the W boson, the scale at which the weak interaction becomes confined.

Scale	Compactness $\mathcal{C}_S(r)$	Horizon radius	Value
Gravitational	$\frac{GM}{c^2 r}$	$r_S = \frac{2GM}{c^2}$	variable
Electromagnetic	$\frac{e^2}{4\pi\epsilon_0 m_e c^2 r}$	$r_e^{\text{cl}} = \frac{e^2}{2\pi\epsilon_0 m_e c^2}$	2.82 fm
QCD	$\frac{4\alpha_s}{3} \cdot \frac{\hbar c}{\Lambda_{\text{QCD}} r}$	$r_p = \frac{8\alpha_s \hbar c}{3\Lambda_{\text{QCD}}}$	0.843 fm
Higgs	$\mathcal{C}_H(r)$ (see text)	$r_e^H = \frac{e^2}{8\pi\epsilon_0 m_e c^2}$	2.82 fm
Electroweak	$\frac{g_W^2 \hbar c}{\mathcal{H}_W^{(0)} r}$	$r_W = \frac{\hbar}{m_W c}$	2.5×10^{-3} fm

Remark 2 (The electromagnetic and Higgs horizons coincide). *The table shows that the electromagnetic horizon radius $r_e^{\text{cl}} \approx 2.82$ fm and the Higgs horizon radius $r_e^H \approx 2.82$ fm are numerically identical. This is not a coincidence of units. In AVF, both conditions describe the same physical boundary of the electron: the electromagnetic scale exhausts its vacuum energy at the same radius at which the Higgs domain wall forms. The two scale descriptions are therefore self-consistent at this radius, and their coincidence defines the structure of the electron. The full significance of this self-consistency is developed in the electron structure paper of this series [2].*

7 Conclusion

In this paper, we have derived the universal nonlinear deformation law $\mathcal{H}_S^{\text{eff}} = \mathcal{H}_S^{(0)} \sqrt{1 - 2\mathcal{C}_S}$. It is shown that the Newtonian deformation field gives the linear profile $\mathcal{H}_S^{(0)}(1 - \mathcal{C}_S)$, whose zero-crossing falls at $r_S/2$. However, the two relativistic corrections, namely, the gravitational redshift of the ambient energy and the stretching of the radial escape path, replace the linear depletion fraction $\mathcal{C}_S$ with the nonlinear form $1 - \sqrt{1 - 2\mathcal{C}_S}$. The effective ambient energy density at scale S is then $\mathcal{H}_S^{\text{eff}} = \mathcal{H}_S^{(0)} \sqrt{1 - 2\mathcal{C}_S}$, whose zero-crossing falls exactly at $\mathcal{C}_S = 1/2$. This is the horizon condition.

The law applies at every energy scale with the compactness $\mathcal{C}_S(r)$ appropriate to that scale. At the gravitational scale it gives the Schwarzschild radius $r_S = 2GM/c^2$. At the QCD scale it gives the proton confinement radius $r_p = 0.843$ fm. At the Higgs scale it gives the electron membrane radius $r_e = 2.82$ fm. The force at each scale is the gradient of the deformation field in the appropriate limit. The non-relativistic limit recovers the classical long-range force laws: Newton's gravity [8], Coulomb's electrostatics [7], and the onset of QCD confinement in their respective domains.

The single law and the same horizon condition, derived from the same pair of relativistic corrections, govern vacuum depletion from galactic scales down to the quark confinement scale. The electromagnetic and Higgs horizon radii are numerically identical at ≈ 2.82 fm; the physical interpretation of this coincidence in terms of electron structure is developed in the companion paper on electron structure [2].

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